



CESAB
CENTRE DE SYNTHÈSE ET D'ANALYSE
SUR LA BIODIVERSITÉ

INTRODUCTION AUX MÉTA-ANALYSES ET REVUES SYSTÉMATIQUES

Critical appraisal of study validity

Joseph LANGRIDGE – Chargé de mission scientifique «
Revue systématique et synthèses »

SOUTIENS INSTITUTIONNELS



PLAN

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When & why do a critical appraisal

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Some founding principles & recurring problems

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Applying FAIR principles

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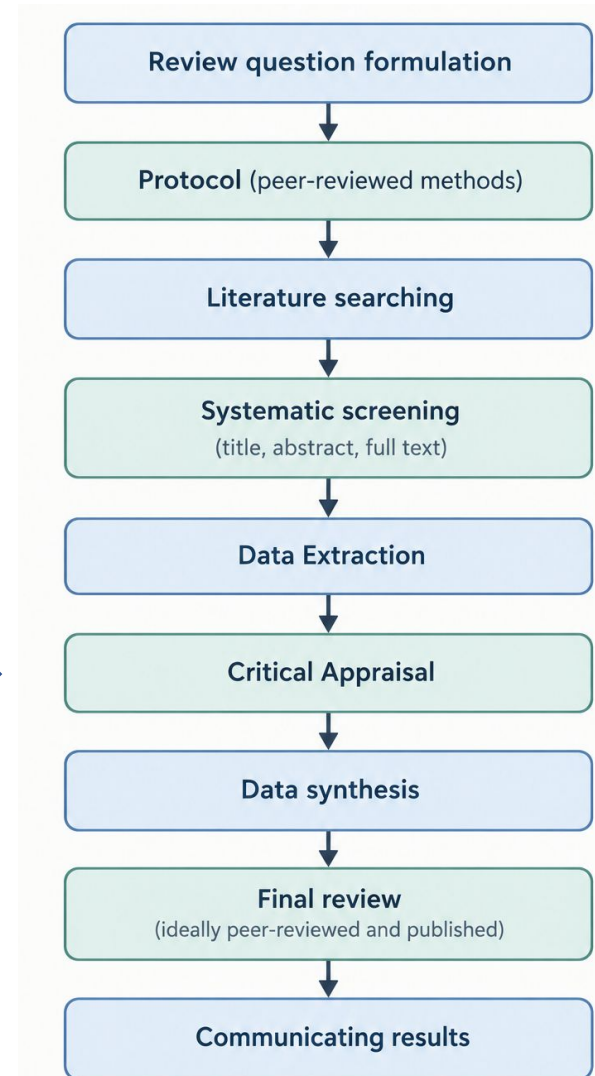
CEE critical appraisal tool

5

Other tools

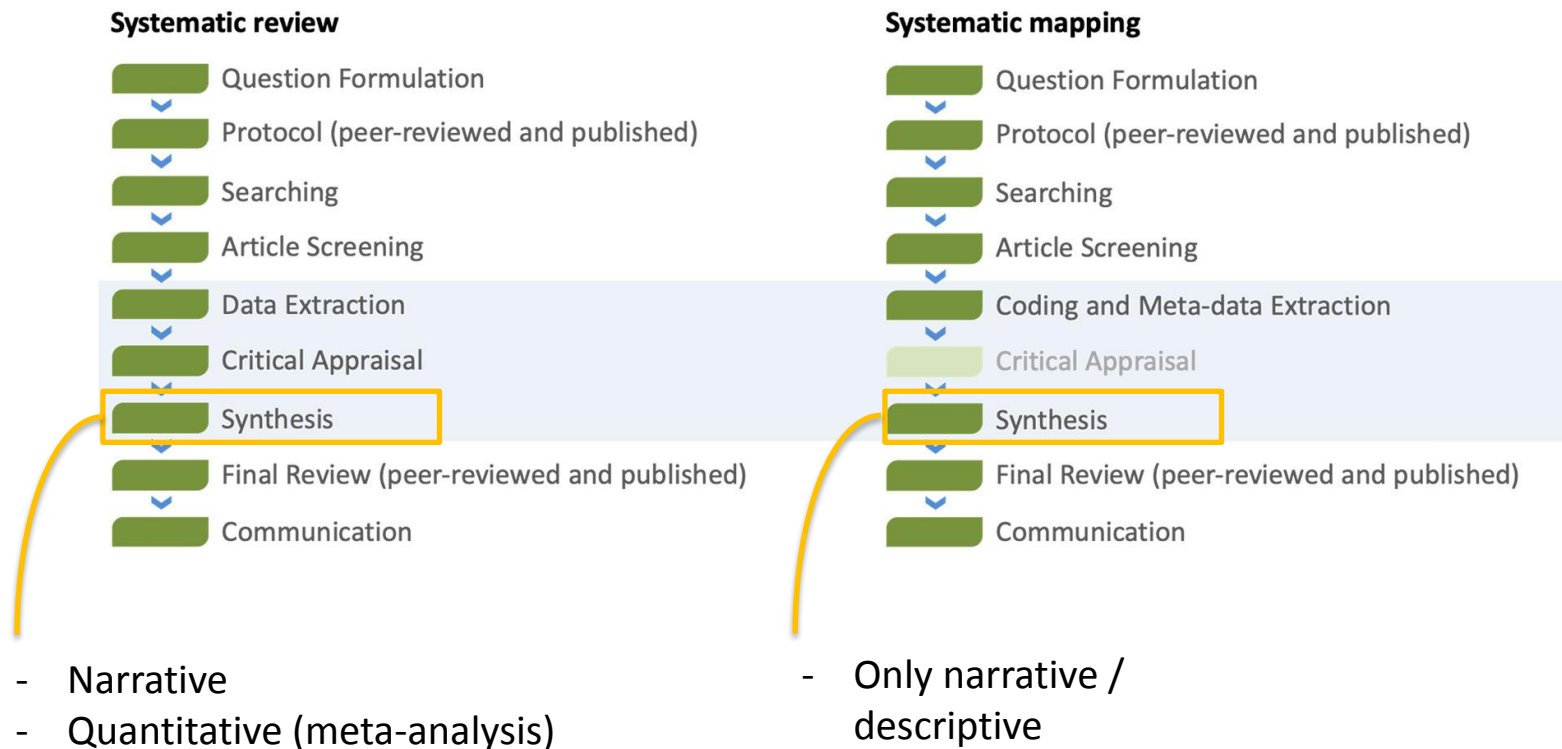
Where/when to undertake critical appraisal?

...if you are undertaking a Systematic review or Meta-analysis, critical appraisal is **mandatory**



Where/when to undertake critical appraisal?

Mandatory for quantitative syntheses



Source: Haddaway et al. 2020. *Campbell Systematic Reviews*. <https://doi.org/10.1002/cl2.1129>

Where/when to undertake critical appraisal?

Mandatory for quantitative syntheses

- Critical appraisal is carried out after the various screening stages: on the **final corpus**.
 - ✓ We now know all the articles that deal with the given question (PICO/PECO).
- Critical appraisal focuses on the ~~quality~~ **validity** of the studies.
- Critical appraisal is generally carried out on **studies/study units** (prior breakdown of “articles” ==> “study units”).
 - ✓ It is possible to **group together** several studies from the same publication if their experimental protocol is the same

Source: Haddaway et al. 2020. *Campbell Systematic Reviews*. <https://doi.org/10.1002/cl2.1129>



Why do a critical appraisal?

Founding principles

- **Key founding point 1:**

True replications of an original observation or experiment are **uncommon** in the scientific literature. This is largely because research funding structures and publication incentives favour novel contributions over efforts to verify existing findings.

- **Key founding point 2:**

Conducting “*perfect*” **methodologically flawless research** is almost non-existent, due to constraints on time and resources — a challenge that becomes even more pronounced in fieldwork, where conditions cannot be controlled as they can in a lab.

Why do a critical appraisal?



Outcome1	Outcome2
Increase	28.9
No effect	2
Increase	11.41

Comparison code	Biblio ID	Biblio author(s)	Biblio Year	Biblio title
Leski_2019_1	35	Leski, T; Rudawska, M	2019	Both forest reserves and ma
Petzold_2018_1	268	Petzold, J; Dittrich, S	2018	Effects of forest managemen
Baran_2018_1	151	Baran, J; Pielech, R; B	2018	No difference in plant speci
Horvat_2017_1	302	Horvat, V; Heras, P; G	2017	Intensive forest managemen
Dvorak_2017_1	345	Dvorak, D; Vasutova,	2017	Macrofungal diversity patter
Horvat_2017b_1	370	Horvat, V; Biurrun, I;	2017	Herb layer in silver fir - beed

Outcome1	Outcome2
Increase	28.9
Increase	4.63
No effect	2
Increase	11.41
Increase	16.07
Increase	26.84



Why do a critical appraisal?

Interpretating forest plots

12 study units (12 articles)
PICO ou PECO

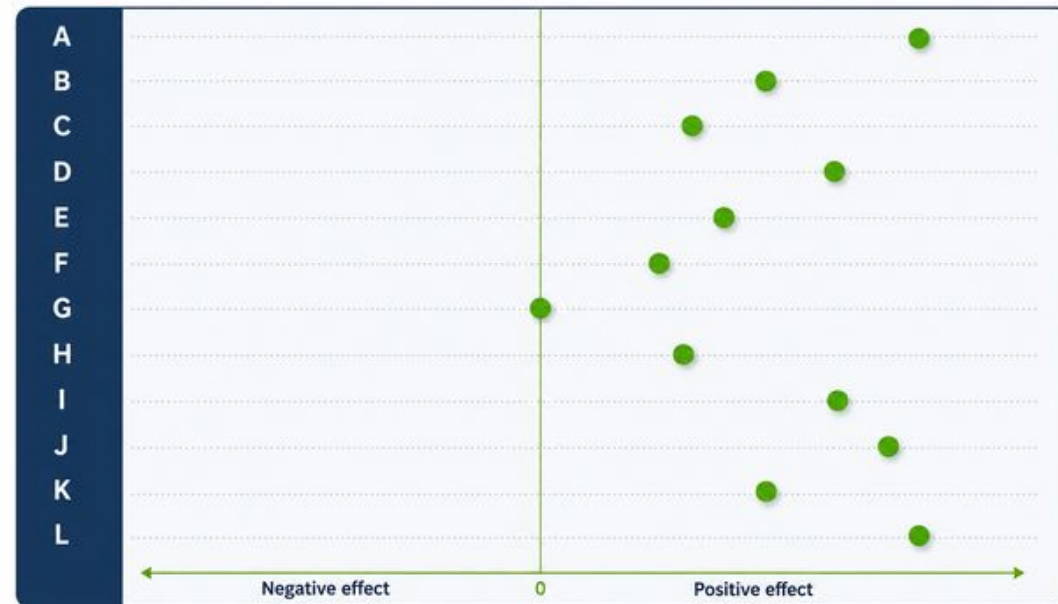


We choose an outcome (metric)



Produce a « forest plot »

Example 1: consensus?

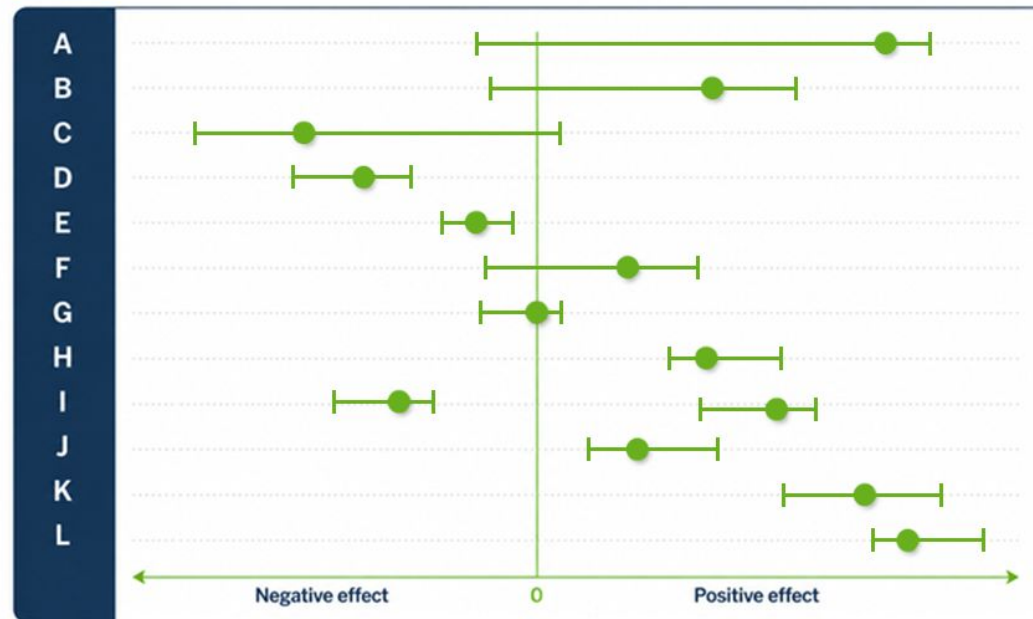


Source: adapted from Livoreil, B. IRN Wan@bi - Atelier2023

Why do a critical appraisal?

Interpretating forest plots

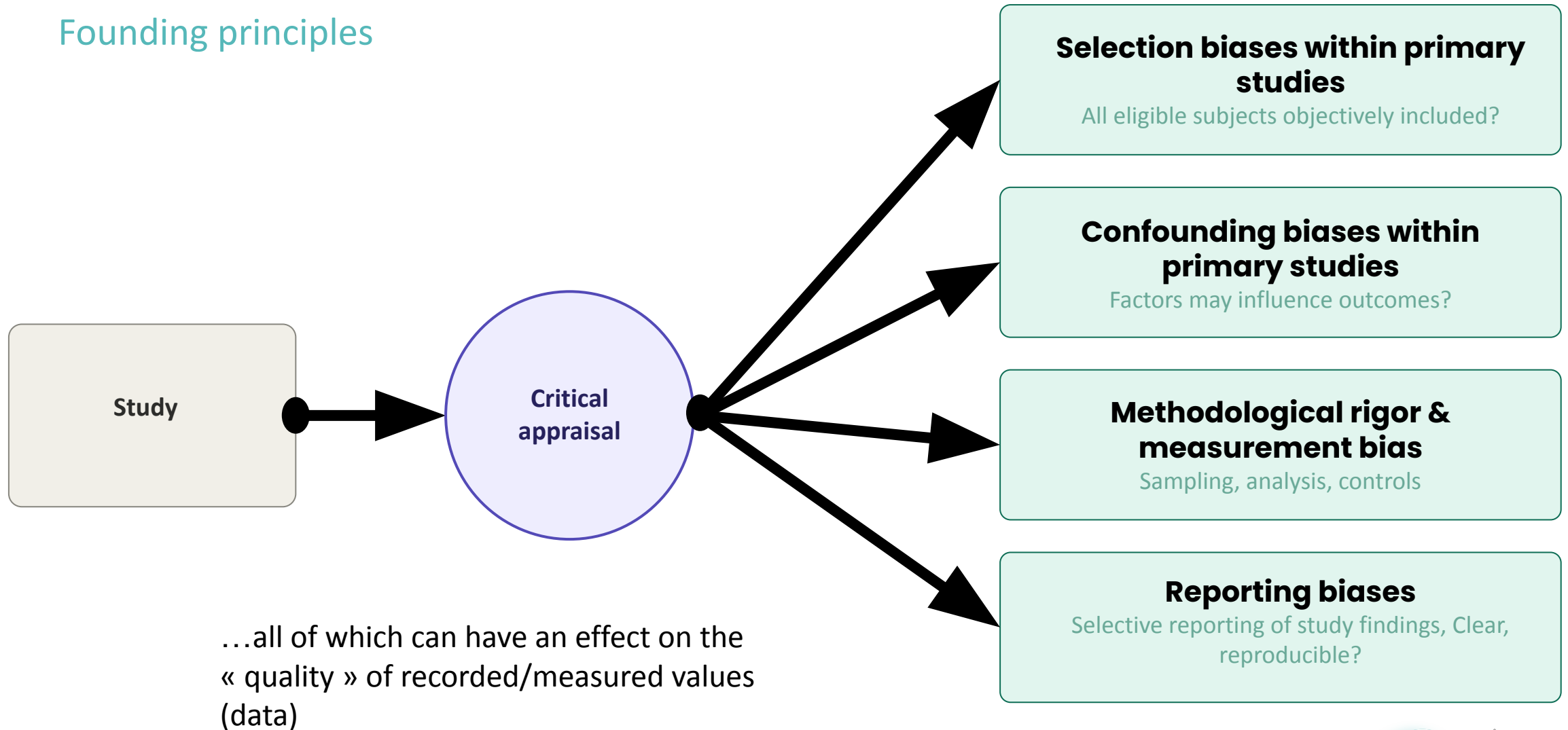
Example 2: heterogeneity?



Source: adapted from Livoreil, B. IRN Wan@bi - Atelier2023

Why do a critical appraisal?

Founding principles



Why do a critical appraisal?

Random vs systematic error in primary studies

- **Random error** = imprecise (false) measurement/observation due to an accident (chance) or a temporary defect. **Unintentional**.

If measures are repeated, a good chance that it **will not be** reproduced.

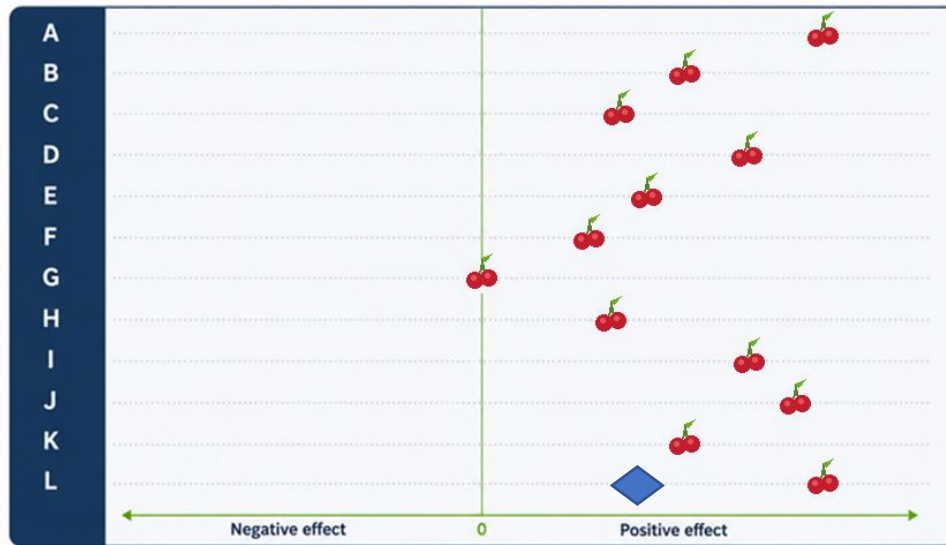
Bias = systematic error. It is not erased by repetition and distorts reality. It can sometimes be “voluntary”.

If systematic error is known but not controlled. Leading to an incorrect answer.

Why do a critical appraisal?

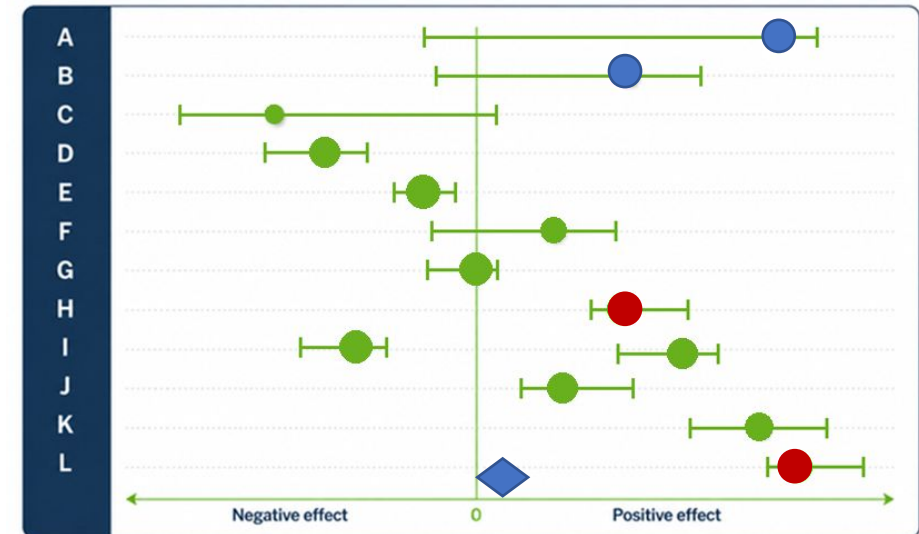
Recurring problems

Example 1: consensus?



Cherry picking

Example 2: heterogeneity?



Biased synthesis

Bias = is that number actually correct? It's driven by things like *confounding factors, selection bias, measurement error, or flawed methodology.*

An example of a biased synthesis

INSECT POPULATIONS

Meta-analysis reveals declines in terrestrial but increases in freshwater insect abundances

Roel van Klink^{1,2,3*}, Diana E. Bowler^{1,4,5}, Konstantin B. Gongalsky^{6,7}, Ann B. Swengel⁸,
Alessandro Gentile¹, Jonathan M. Chase^{1,9}

Comment on “Meta-analysis reveals declines in terrestrial but increases in freshwater insect abundances”

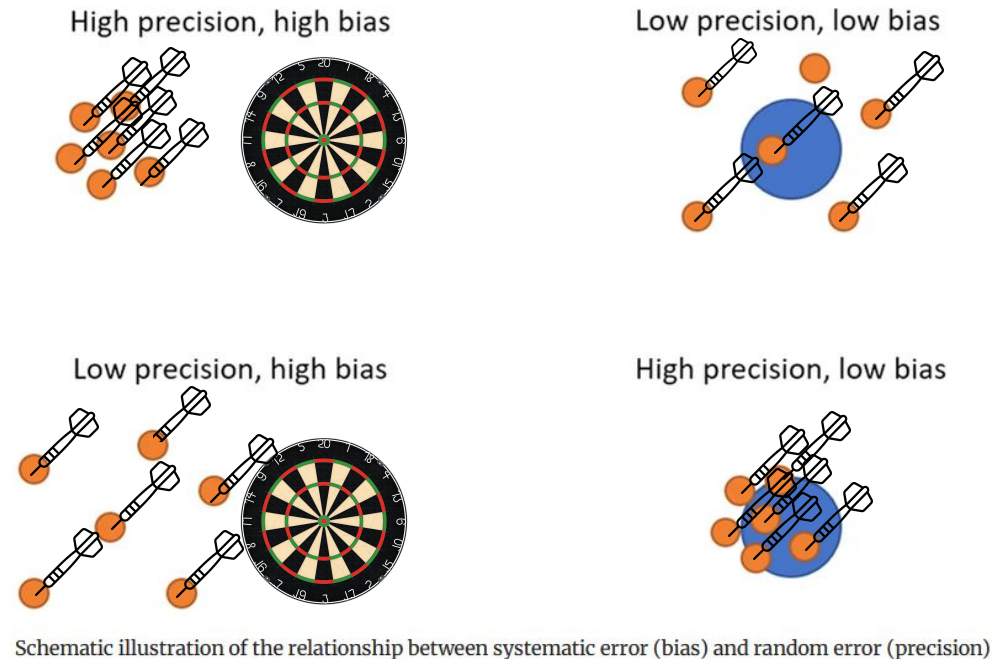
Marion Desquilbet^{1*†}, Laurence Gaume^{2†}, Manuela Grippa³, Régis Céréghino⁴,
Jean-François Humbert⁵, Jean-Marc Bonmatin⁶, Pierre-André Cornillon⁷, Dirk Maes⁸,
Hans Van Dyck⁹, David Goulson¹⁰

<https://www.science.org/doi/10.1126/science.abd8947>

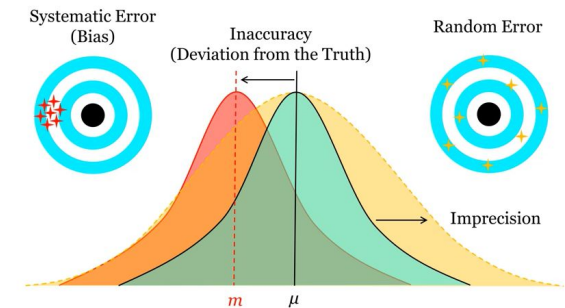
- No attempt was made to weight studies
- Confounding factors
 - geographic location,
 - anthropogenic impact (including farming methods and pesticide use),
 - protected status

The dartboard analogy

- This is what makes it dangerous in evidence synthesis: most meta-analyses use inverse-variance weighting, meaning **more precise studies get more weight in the pooled estimate** — precision, not validity, drives influence. If a biased study happens to also be large and precise, it can dominate the pooled result and drag the whole synthesis toward its systematic error, while genuinely rigorous but smaller studies get diluted.



Systematic Error Vs. Random Error



Source: Frampton, *et al.* 2022. *Environ Evid.* <https://doi.org/10.1186/s13750-022-00264-0>



FAIR principles

Focused: select appropriate and specific validity constructs (i.e. internal validity or external validity).

Extensive: capture all aspects of the validity construct (i.e. if the construct is internal validity, all the different types of bias that could arise in a given study design must be identified and assessed).

Applied: inform the data synthesis step of the Systematic review or Meta-analysis in an appropriate way.

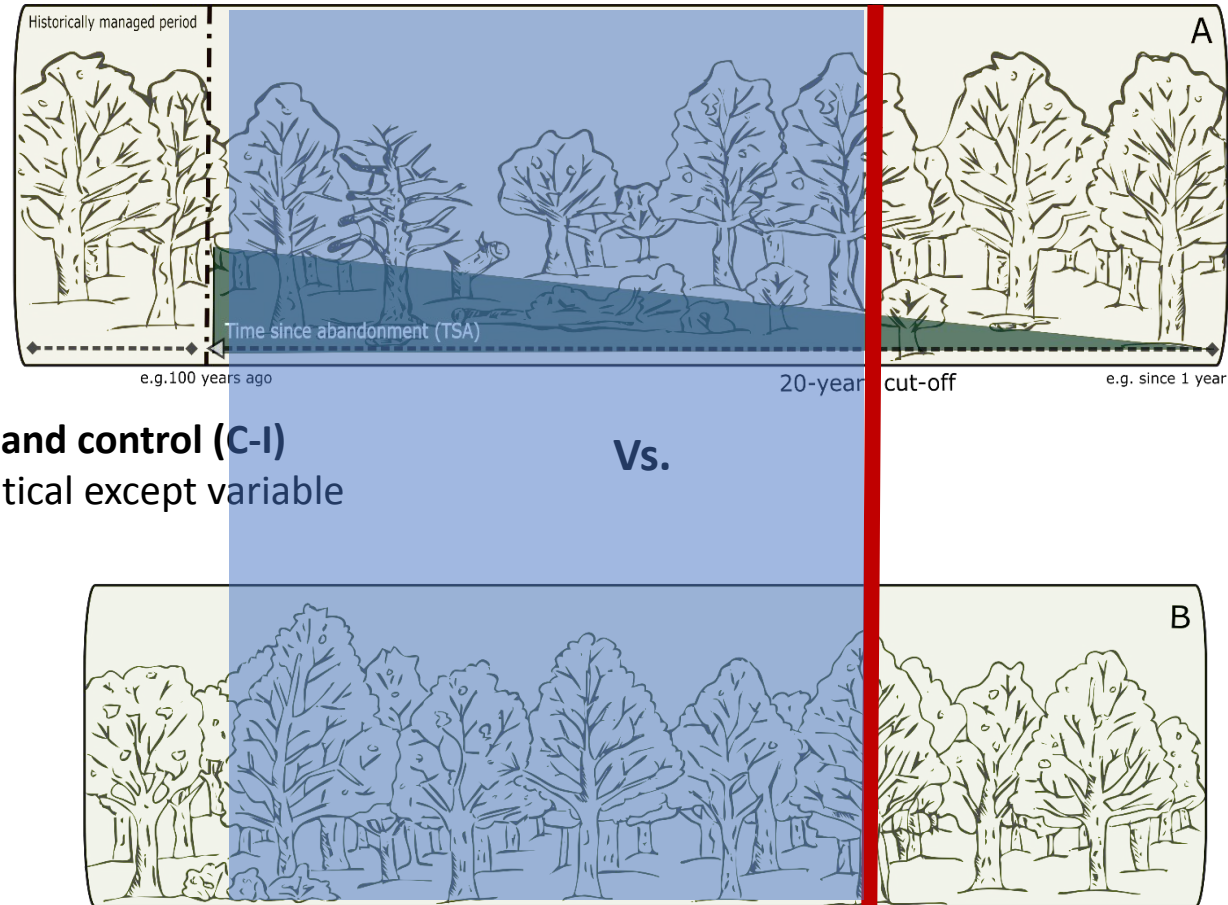
Transparent: maximise objectivity and clarity.

Applying FAIR

Focused

Intervention and control (C-I)
sites are identical except variable
of interest

Vs.

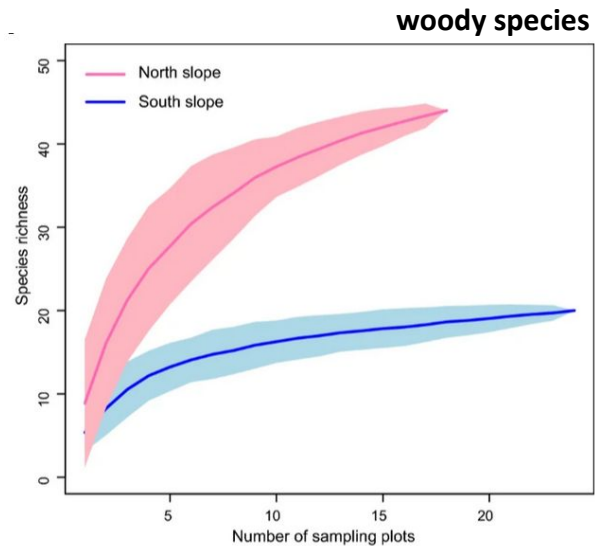


Site type
Gross successional stages
Sampling design / comparator

Langridge *et al.* 2023. *Biological Conservation*. <https://doi.org/10.1016/j.biocon.2023.110296>

Applying FAIR

Extensive



Yang et al. 2020. *Scientific reports*.

Site type

Topography

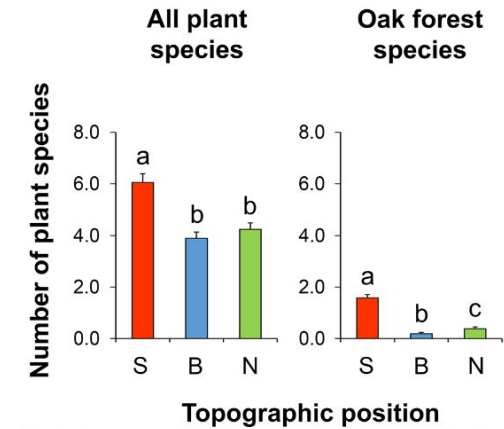


Fig. 5 Number of vascular plant species (all plant species and oak forest species) in the plots of topographic positions in dolines (S: south-facing slope, B: bottom, and N: north-facing slope) (mean $\pm$ SE). Significant differences ($p < 0.05$, based on the GLMMs with Bonferroni post hoc tests) are indicated by different lower case (a–c) letters

Bátori et al. 2023 *Annals of Forest Science*.

Principle:

South vs North



North unmanaged vs north managed

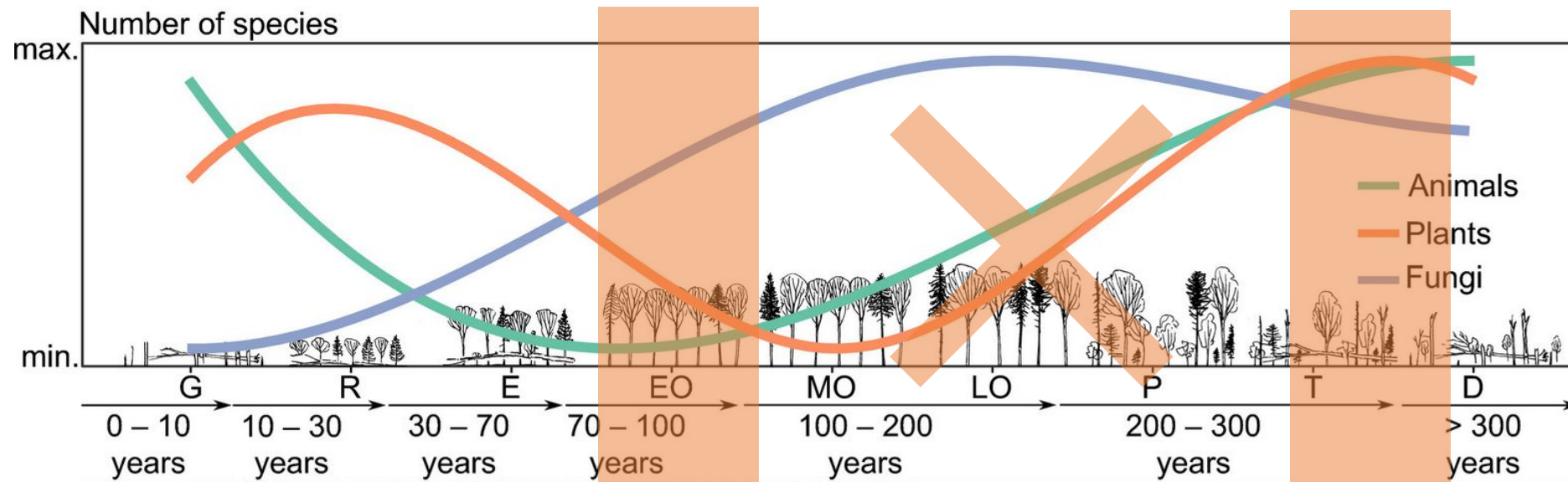
South unmanaged vs south managed



Langridge et al. 2023. *Biological Conservation*. <https://doi.org/10.1016/j.biocon.2023.110296>

Applying FAIR

Successional stages

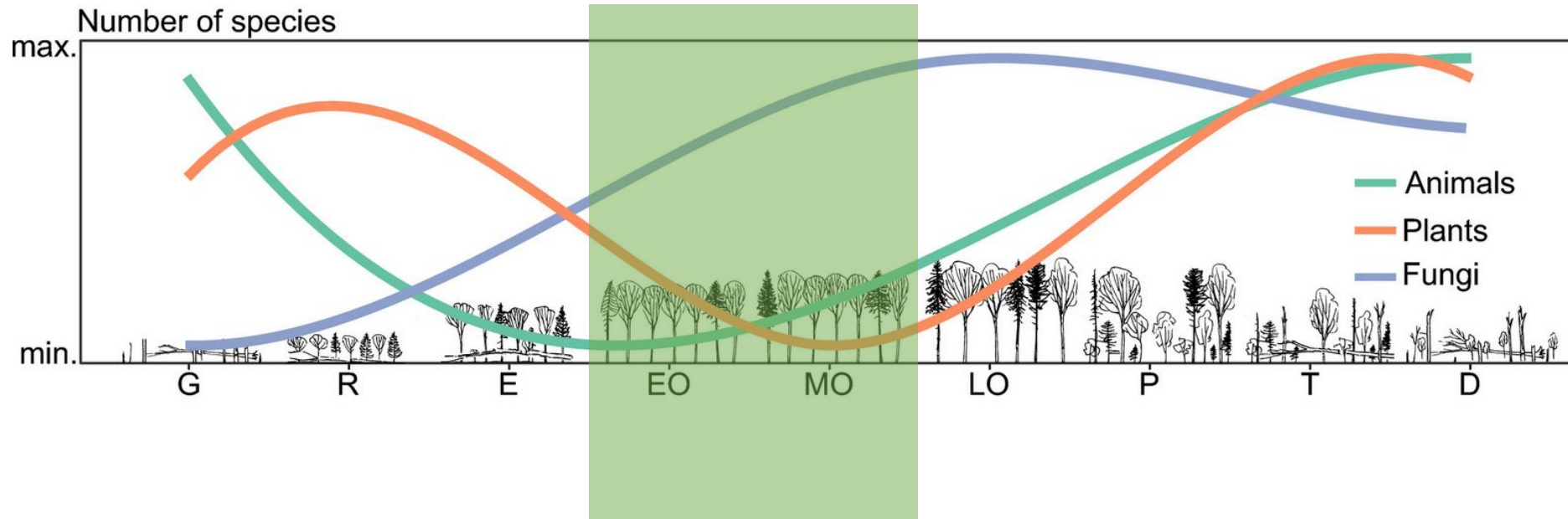


G, gap; R, regeneration; E, establishment; EO, early optimum; MO, mid-optimum; LO, late optimum; P, plenter; T, terminal; D, decay

Hilmers et al. 2018. *Journal of Applied Ecology*. <https://doi.org/10.1111/1365-2664.13238>

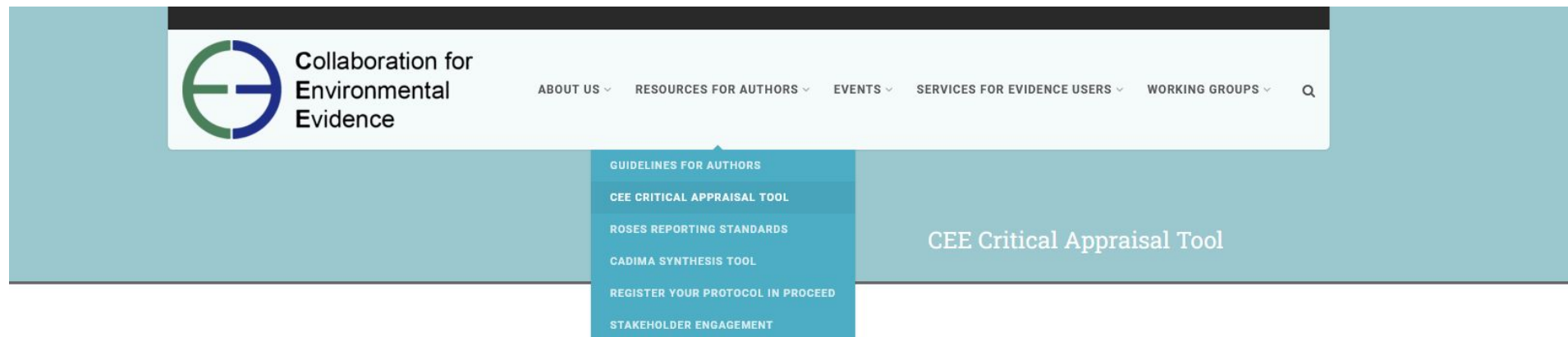
Applying FAIR

Sampling design / comparator



EO managed vs EO unmanaged

CEE critical appraisal tool



Collaboration for Environmental Evidence Critical Appraisal Tool Version 0.3 (Prototype)

Last updated: 24 October 2021

PROJECT SUMMARY

We are currently developing a critical appraisal tool for evaluating 'risk of bias' (or threats to internal validity) of primary studies assessing effectiveness of interventions or impacts of exposures in environmental management. There are well-known, widely applied risk of bias assessment tools in the health sector known as 'RoB 2' and 'ROBINS-I' (www.riskofbias.info), but there are currently no such critical appraisal tools in environmental management. Here we provide a third draft of the tool. The tool is still under development and requires initial testing, but it may help environmental evidence synthesists conduct critical appraisal. In the latest version, more descriptions are provided in Part A, and revisions and clarifications are made in Part B. We have merged two risk-of-bias criteria and so the total number of risk-of-bias criteria is reduced to seven in version 0.3.

If you are not familiar with critical appraisal step, we recommend reading Sections 3.5 (www.environmentalevidence.org/guidelines/section-3) and 8 (www.environmentalevidence.org/guidelines/section-8) of *CEE Guidelines* before you download the

Resources for Authors

Guidelines for Authors

- [Aims and Scope](#)
- [Table of Contents](#)
- [Updates and Corrections](#)
- [1. Process Summary](#)
- [2. Need for Evidence, Synthesis Type and Review Team](#)
- [3. Planning a CEE Evidence Synthesis](#)
- [4. Writing and Registering a Protocol](#)
- [5. Conducting a Search](#)
- [6. Eligibility Screening](#)
- [7. Data Coding and Data Extraction](#)
- [8. Critical appraisal of study validity \(SRs\)](#)
- [9. Data Synthesis](#)
- [10. Interpreting findings and reporting](#)



Scope of tool

Within the scope

☐ If all of the followings apply

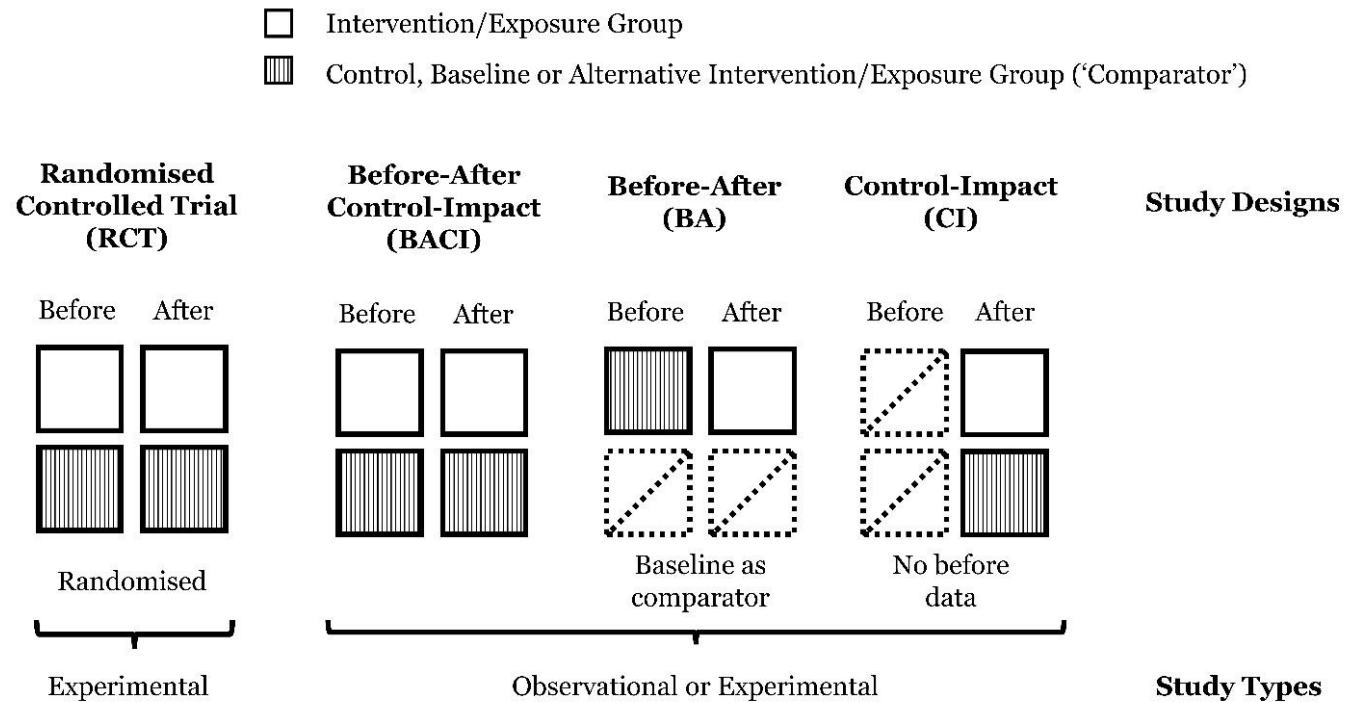
- ☐ Review question justifiably relates to environmental management (policy or practice)
- ☐ Reviewing evidence on impact of exposure or effectiveness of intervention
- ☐ Interest is quantitatively measured outcomes

Beyond the scope

☐ If any of the followings applies

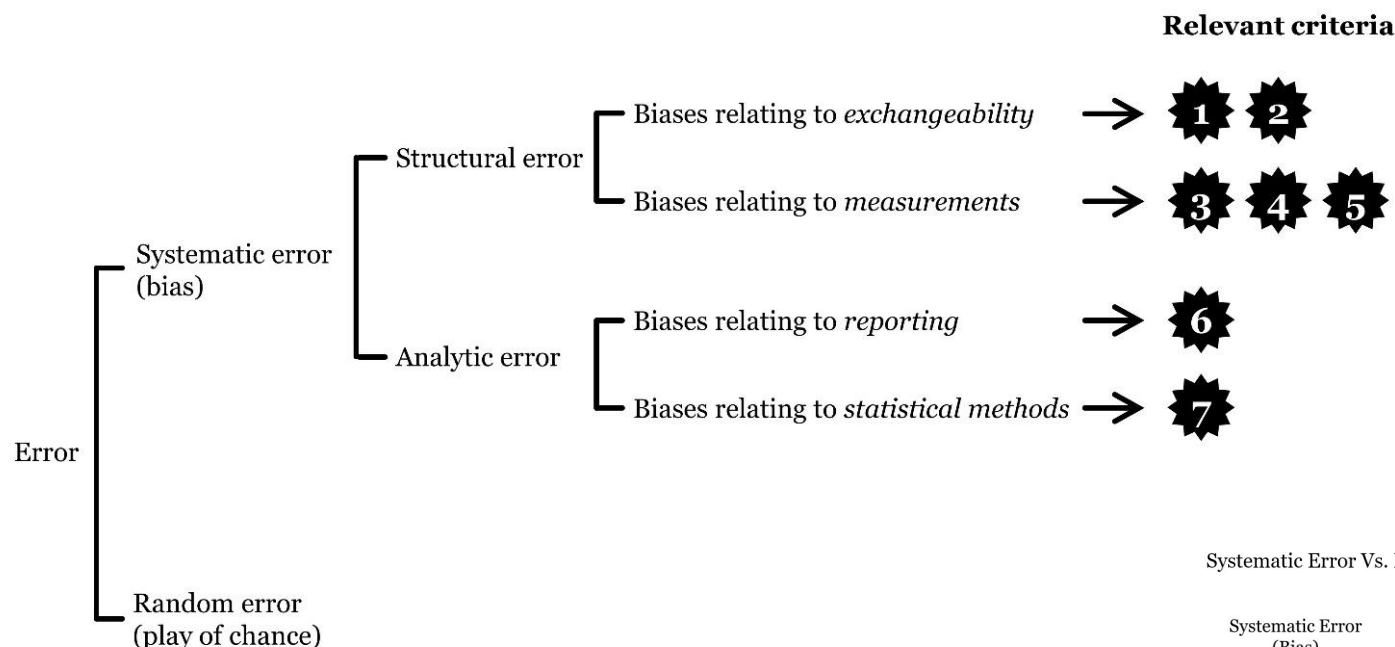
- ☐ Reviewing medical research involving human subjects, tissues, or personal data (including physiological, biomechanical, psychological research)
- ☐ Reviewing purely laboratory-based biological research (e.g., *in vitro* or *in vivo* experiments, genome sequencing)
- ☐ Reviewing qualitative evidence

Assumptions of the tool



- ✓ The tool does not assume that study types are biased.
- ✓ The tool does not assume the differences in study design alone affect precision.

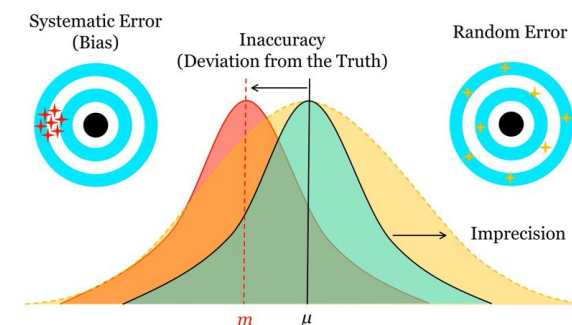
Organisational Schema for errors



Structural error: bias relating to measurement of **intervention, exposure or outcome**, and bias relating to **exchangeability**.

Exchangeability refers to **independence** between the outcome and the observed intervention.

Systematic Error Vs. Random Error



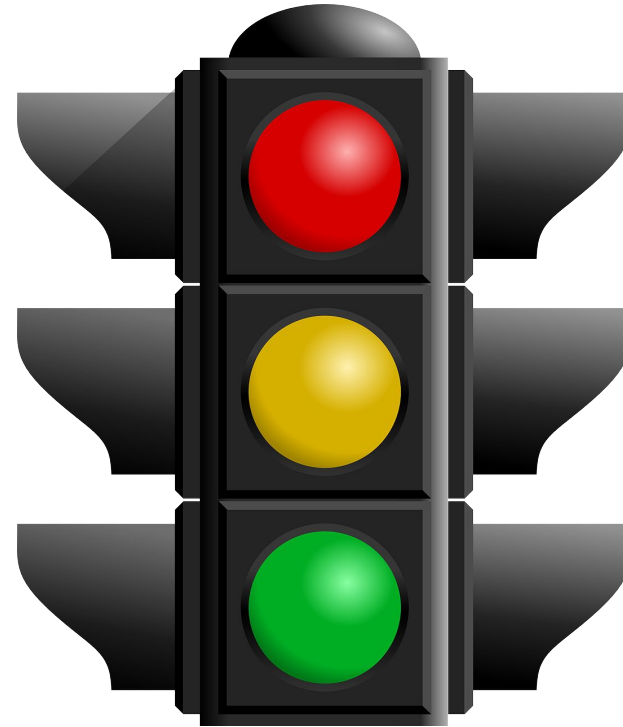


Risk of bias judgement

Within Individual Criteria

Three levels:

- Low risk of bias (**Low**)
- Medium risk of bias (**Med**)
- High risk of bias (**High**)



Risk of bias classes

Summary

Source: Frampton, et al. 2022. *Environ Evid.* <https://doi.org/10.1186/s13750-022-00264-0>
Konno et al. 2024. *Environ Evid.* <https://doi.org/10.1186/s13750-024-00324-7>

1. Bias due to confounding factors **“Risk of confounding biases” in the CEE tool.** One or more uncontrolled variables (**confounders**) that **influence** both the **exposure** and the **outcome**.
2. Bias in selection of subjects/areas into the study **“Risk of post-intervention/exposure selection biases” in the CEE tool.** When eligible subjects are excluded in a way that leads to a false association between the exposure and outcome.
 - 2a. Bias due to missing data (attrition bias) Bias due to **missing data** can be considered as a type of selection bias. When follow-up data of subjects that are initially included in the study **are not fully available** for inclusion in the analysis. Thus, an **imbalance** in the **amount of missing data** between the **exposure/intervention** and **comparator groups**.
3. Bias due to misclassification of the exposure **“Risk of misclassified comparison biases”.** Misclassification or mismeasurement of the exposure and/or comparator leads to a **misrepresentation** of the **association** between the **exposure** and the **outcome** (*also known as measurement bias or information bias*).
4. Bias due to deviation from the planned exposure (intervention) in experimental studies **“Risk of performance biases”.** **Alteration** of the **planned exposure** or **comparator** treatment procedure(s) of interest **after the start** of the exposure.

Risk of bias classes

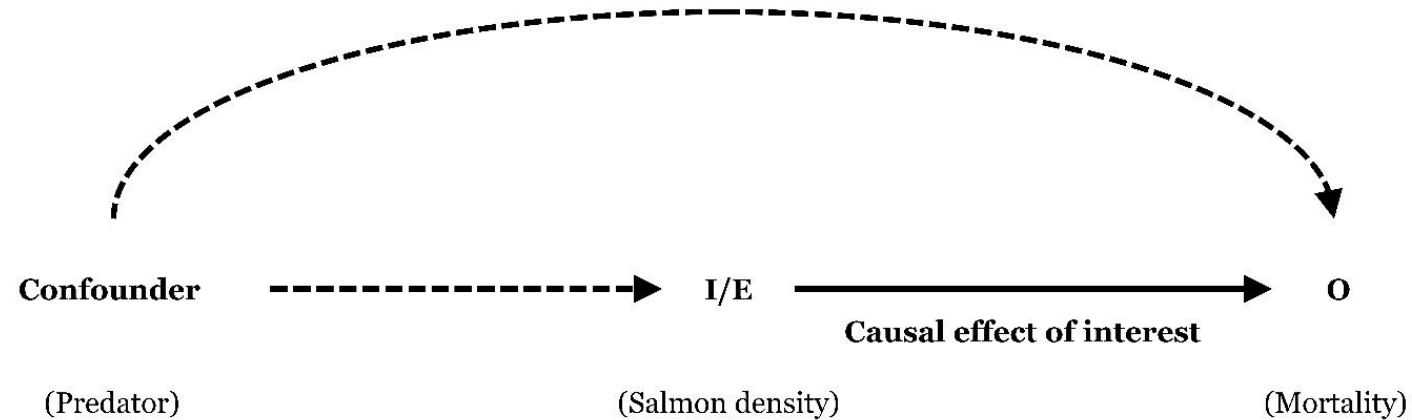
Summary

- 5. Bias in measurement of outcomes** **“Risk of detection biases”**. **Systematic differences** in measurements of **outcomes**. **Systematic errors** in measurement of **outcomes** may occur if outcome data are determined differently between the exposure and comparator groups. This could be intentional (e.g. influence of desire to obtain a certain direction of effect) or unintentional (e.g. due to cognitive bias or human errors).
- 6. Bias in selection of the reported results** **“Risk of outcome reporting biases”**. Selective reporting of study findings. Due to (i) presentation of selected findings from multiple measurements; (ii) presentation of results for selected subgroups or subpopulations of the planned analysis population; and (iii) presentation of selective findings from multiple analyses.
- 7. Bias due to an inappropriate statistical analysis approach** **“Risk of outcome assessment biases”**. Errors in statistical methods applied in the primary studies. Areas to check:
- (i) **errors in applied descriptive statistics** (e.g. miscalculation of sample sizes, means, or variances, including pseudoreplication);
 - (ii) **errors in applied inferential statistics** (including flawed null hypothesis testing, estimation, or coding);
 - (iii) **use of inappropriate statistical tests** or **violation of assumptions** required by tests (e.g. criteria for normality and equal variances are not satisfied)

Risk of confounding biases (CEE tool)

Criterion 1

Risk of Confounding Biases



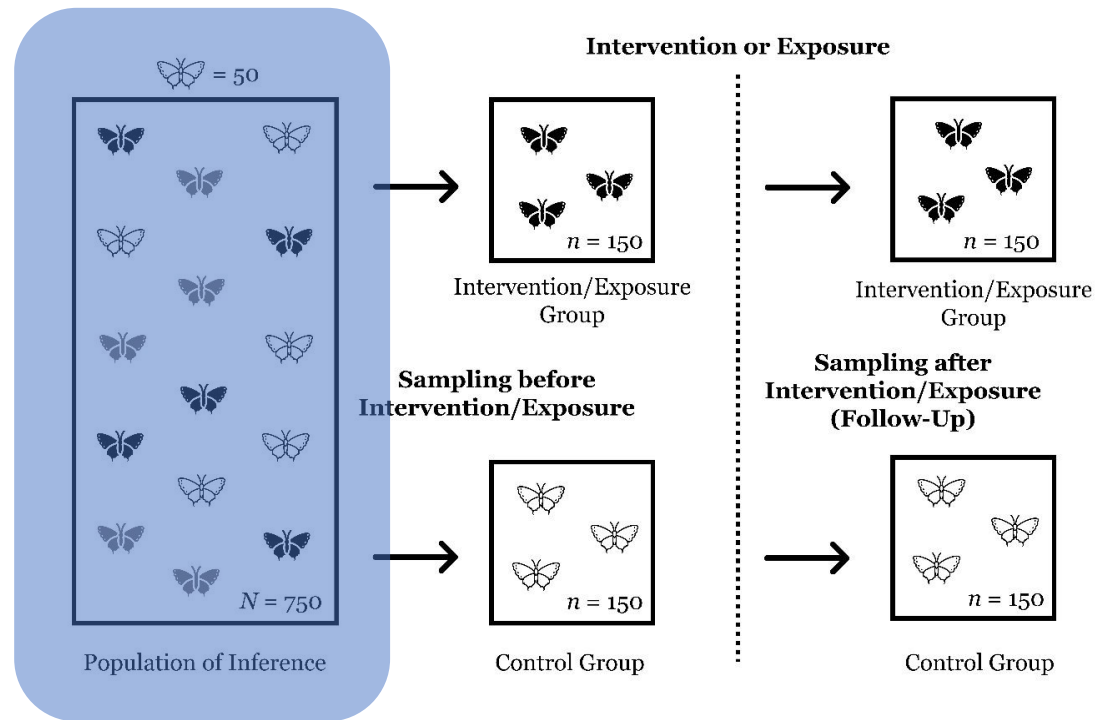
Understand the effect of salmon density on their survival?

Presence of predators ☐ affect rate of survival (O) thus density (I/E).

Risk of confounding biases (CEE tool)

Criterion 1

Risk of Confounding Biases



Need to understand
variability

e.g. morphological
differences (covariate)
may explain the effect



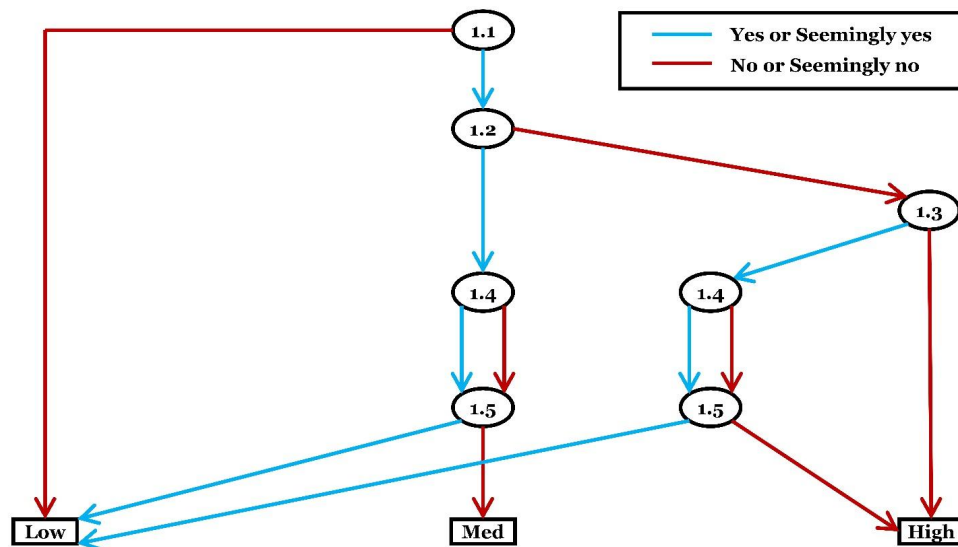
Stratified sampling
Divided into
sub-populations

Risk of confounding biases (CEE tool)

Criterion 1



Questions already developed by leading experts



1.1. Is confounding possible?

1.2. Are the potential confounders controlled for?

1.3. Is there any justification for not controlling for the potential confounders?

1.4. Are the potential confounders likely to be measured accurately and precisely?

1.5. Are the analyses of the effect appropriate?

Tool template ready to go



Criterion 1: Risk of confounding biases			
Category	Checklist questions	Response option	Answer
General (please answer)	1.1. Is it possible for the impact of the exposure or the effectiveness of the intervention to be confounded in this study?	Yes Seemingly yes Seemingly no No	
Conditional (answer if Y/SY to 1.1, otherwise select 'Not applicable')	1.2. Did the author(s) control for all the potential confounders?	Yes Seemingly yes Seemingly no No Unclear (No) Not applicable	
Conditional (answer if N/SN/Unclear to 1.2, otherwise select 'Not applicable')	1.3. Is there any justifiable reason for not controlling for all the potential confounders (so that omission of some of the potential confounders is unlikely to influence the assessment of the effectiveness or impact)?	Yes Seemingly yes Seemingly no No Not applicable	

▶
Citation
Criterion1
Criterion2
Criterion3
Criterion4
Criterion5
Criterion6
Criterion7
Overall
All_decisions
+

Source: Konno K, Livoreil B, Pullin AS. 2021. Collaboration for Environmental Evidence Critical Appraisal Tool version 0.3.
<https://environmentalevidence.org/cee-critical-appraisal-tool/>

Other available tools at the ready



About Resources and training Methods in Cochrane Join Cochrane

Risk of Bias 2 (RoB 2) tool

- Process for proposing changes to methods or tools used in Cochrane
- Methods Support Unit
- Contact Methods Support or Methods Groups
- Clinical study reports and other regulatory documents
- Data-based predictive distributions for between-study heterogeneity
- Repeated meta-analyses
- Risk of Bias 2 (RoB 2) tool

The Risk of Bias 2 (RoB 2) tool is an update to the original risk of bias tool that launched in 2008. The relevant chapter in the *Cochrane Handbook for Systematic Reviews of Interventions* Chapter 8, titled 'Assessing risk of bias in a randomized trial'. The Methodological Expectations for Cochrane Intervention Reviews (MECIR) Manual includes standards for assessing risk of bias in included studies; C52-60. Up-to-date information from the developers on RoB 2 is available via the Risk of Bias tools website: www.riskofbias.info.

Key Cochrane resources for using RoB 2 in Cochrane Reviews are:

[An Introduction to Risk of Bias 2](#)

<https://methods.cochrane.org/bias/resources/rob-2-revised-cochrane-risk-bias-tool-randomized-trials>



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Checklist for Systematic Reviews

How to cite



Aromataris E, Fernandez R, Godfrey C, Holly C, Kahlil H, Tungpunkom P. Summarizing systematic reviews: methodological development, conduct and reporting of an Umbrella review approach. *Int J Evid Based Healthc*. 2015;13(3):132-40.

Associated publication(s)



JBI Manual for Evidence Synthesis
Chapter 10: Umbrella Reviews

<https://jbi.global/critical-appraisal-tools>

Campbell Collaboration

The Campbell Collaboration is an international research network that produces systematic reviews of the effects of social interventions in Crime & Justice, Education, International Development, and Social Welfare.

[BACK TO GLOSSARY](#)

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MESSAGES CLÉS

- 1 | It must be carried out if you are undertaking a quantitative synthesis : No excuses!
- 2 | Be FAIR
- 3 | Choose the right tool, there are many available ranging from checklist-style tools to in-depth appraisal
- 4 | Anticipate the time needed to undertake critical appraisal, adapt your project objectives accordingly